Progressive exercises

E1 · Practice. Determine $\lim\limits_{x\to+\infty} \dfrac{5x^3 - x}{2x^3 + 7}$, with justification.

E2 · Practice. Let f be the function defined on $\mathbb{R} \setminus \{2\}$ by $f(x) = \frac{1}{(x-2)^2}$. Determine its limits at 2^-, at 2^+, at $-\infty$ and at $+\infty$.

E3 · Practice. Determine $\lim\limits_{x\to1} \dfrac{x^2 - 1}{x - 1}$, stating why the simplification used is valid in a neighbourhood of 1.

E4 · Practice. Determine $\lim\limits_{x\to+\infty} x^3 e^{-2x}$. State the result on comparison of growth rates that is used.

E5 · Practice. Determine $\lim\limits_{x\to-\infty} \left(\sqrt{x^2+1} + x\right)$. Justify in particular the treatment of $\sqrt{x^2}$.

Textbook or past-paper work

Barbazo, p. 96, exercises 26 and 29. In S23: exercise 29, to link limits and asymptotes. At home after S24: exercise 26, to construct three counterexamples to an invalid rule for calculating with limits.